Residual Asset Burden and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Residual Asset Burden (RAB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on RAB achieves an annualized gross (net) Sharpe ratio of 0.39 (0.35), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 25 (22) bps/month with a t-statistic of 3.58 (3.24), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Book to market, original (Stattman 1980), Book leverage (annual), Operating leverage, Equity Duration, Earnings-to-Price Ratio, Enterprise Multiple) is 21 bps/month with a t-statistic of 3.02.

1 Introduction

Asset prices reflect the compensation investors require for bearing systematic consumption risk, with cross-sectional variation in expected returns arising from differential exposures to aggregate consumption fluctuations. While the consumption-based asset pricing paradigm provides a coherent theoretical framework for understanding return predictability, empirical identification of firm characteristics that capture consumption risk exposure remains challenging. The disconnect between consumption-based theory and cross-sectional return patterns motivates continued investigation into accounting-based measures that may proxy for firms' fundamental risk characteristics.

We hypothesize that Residual Asset Burden (RAB) captures firms' exposure to consumption risk through its reflection of capital intensity and operating inflexibility. Firms with high RAB maintain substantial fixed assets relative to their productive capacity, creating operating leverage that amplifies their sensitivity to aggregate consumption shocks (Carlson et al., 2004). During economic downturns when consumption growth declines, these firms face binding capacity constraints and cannot easily adjust their cost structures, leading to procyclical cash flows that covary strongly with investors' marginal utility of consumption (Zhang, 2005). The consumption-based framework predicts that assets whose payoffs covary positively with consumption command lower risk premiums, as they fail to provide hedging benefits when marginal utility is high (Cochrane et al., 2023). Furthermore, RAB may capture firms' exposure to long-run consumption risks through the duration of their asset base. Firms with high residual asset burdens typically operate with long-lived, specialized capital that embeds expectations about future productivity and consumption growth (Bansal and Yaron, 2004). These firms exhibit greater sensitivity to news about long-run consumption dynamics, as revisions to expected future growth rates have amplified effects on the present value of cash flows from inflexible asset bases. The state-dependent nature of consumption risk further strengthens the predicted relationship between RAB and expected returns. During disaster states or periods of heightened consumption volatility, firms with high RAB face elevated bankruptcy risk due to their operating inflexibility, requiring additional compensation that manifests as higher average returns (Berk et al., 1999).

The RAB trading strategy demonstrates robust return predictability across multiple specifications and risk adjustments. A value-weighted long/short portfolio that buys high RAB stocks and sells low RAB stocks earns an average monthly return of 24 basis points with a t-statistic of 3.06, generating an annualized Sharpe ratio of 0.39. The strategy's alpha remains economically and statistically significant across standard factor models, with the six-factor alpha equaling 25 basis points per month (t-statistic = 3.58). The return predictability of RAB extends beyond small stocks, as evidenced by conditional double sorts on size. Among the largest quintile of stocks by market capitalization, the RAB strategy generates average returns of 19 basis points per month with a t-statistic of 2.17, and six-factor alphas of 22 basis points per month. After accounting for transaction costs using composite bid-ask spreads, the strategy maintains positive and significant net returns of 22 basis points per month with an annualized net Sharpe ratio of 0.35. The RAB signal's predictive power persists after controlling for closely related anomalies. When we simultaneously control for six related characteristics including book-to-market, operating leverage, and equity duration, the RAB strategy retains an alpha of 21 basis points per month with a t-statistic of 3.02.

This paper contributes to the consumption-based asset pricing literature by identifying RAB as a novel characteristic that captures cross-sectional variation in consumption risk exposure. While Fama and French (1993) document that book-to-market ratios predict returns, and subsequent work by Zhang (2005) links this predictability to investment frictions, our RAB measure provides a more direct proxy for operating inflexibility and its associated consumption risk. Unlike traditional leverage measures studied in the accounting literature (Penman et al., 2007), RAB isolates the component of asset intensity orthogonal to standard risk factors, revealing previously unidentified variation in firms' fundamental risk characteristics. Our findings extend the work of Carlson et al. (2004) on operating leverage by demonstrating that residual asset burden captures consumption risk exposure beyond what is explained by conventional operating leverage metrics. Methodologically, we advance the literature by subjecting RAB to comprehensive robustness tests following the protocol of Novy-Marx and Velikov (2023), establishing that the signal's predictive power survives transaction costs and controls for over 200 documented anomalies. The economic magnitude of the RAB premium and its persistence across size groups suggest that consumption-based mechanisms, rather than mispricing or limits to arbitrage, drive the return predictability. These results reinforce the relevance of production-based and consumption-based theories for understanding the cross-section of expected returns, while providing practitioners with an implementable strategy that captures systematic consumption risk exposure.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on industrial firms and exclude financial institutions (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their unique regulatory environments and balance sheet structures.Residual Asset Burden signal requires two key accounting variables from firms' balance sheets. Current assets other sundry (ACOX) represents miscellaneous current assets that do not fit into standard current asset categories such as cash, receivables, or inventory. This item captures residual short-term assets including prepaid expenses, deferred charges, and other sundry current assets not elsewhere classified. Stockholders' equity (SEQ) represents the book value of common equity, calculated as total assets minus total liabilities, reflecting shareholders' residual claim on the firm's assets.construct the Residual Asset Burden signal by dividing current assets other sundry (ACOX) by stockholders' equity (SEQ) for each firm-year observation. This ratio captures the proportion of shareholders' equity tied up in miscellaneous current assets that may represent less productive or harder-to-value components of working capital. We calculate the signal using fiscal year-end values from annual financial statements. To ensure data quality, we require non-missing and positive values for both ACOX and SEQ, and exclude observations where SEQ is less than $1 milliontoavoidextremeratiosdrivenbynear-zerodenominators.Wewinsorizetheresultingratioatthe1stand99thpercentilestomitigatetheinfluence$

3 Signal diagnostics

Figure 1 plots descriptive statistics for the RAB signal. Panel A plots the time-series of the mean, median, and interquartile range for RAB. On average, the cross-sectional mean (median) RAB is 0.06 (0.02) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input RAB data. The signal's interquartile range spans -0.00 to 0.09. Panel B of Figure 1 plots the time-series of the coverage of the RAB signal for the CRSP universe. On average, the RAB signal is available for 6.23% of CRSP names, which on average make up 7.21% of total market capitalization.

4 Does RAB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on RAB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high RAB portfolio and sells the low RAB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short RAB strategy earns an average return of 0.24% per month with a t-statistic of 3.06. The annualized Sharpe ratio of the strategy is 0.39. The alphas range from 0.23% to 0.35% per month and have t-statistics exceeding 2.93 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.39, with a t-statistic of -11.91 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 559 stocks and an average market capitalization of at least $1,551 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the decile sort using NYSE breakpoints and value-weighted portfolios, and equals 12 bps/month with a t-statistics of 1.35. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-two exceed two, and for thirteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 2-24bps/month. The lowest return, (2 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.35. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the RAB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in seventeen cases.

Table 3 provides direct tests for the role size plays in the RAB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and RAB, as well as average returns and alphas for long/short trading RAB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the RAB strategy achieves an average return of 19 bps/month with a t-statistic of 2.17. Among these large cap stocks, the alphas for the RAB strategy relative to the five most common factor models range from 20 to 32 bps/month with t-statistics between 2.26 and 4.15.

5 How does RAB perform relative to the zoo?

Figure 2 puts the performance of RAB in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the RAB strategy falls in the distribution. The RAB strategy's gross (net) Sharpe ratio of 0.39 (0.35) is greater than 81% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the RAB strategy (red line).[2] Ignoring trading costs, a $1 invested in the RAB strategy would have yielded $4.12 which ranks the RAB strategy in the top 8% across the 212 anomalies. Accounting for trading costs, a $1 invested in the RAB strategy would have yielded $3.24 which ranks the RAB strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the RAB relative to those. Panel A shows that the RAB strategy gross alphas fall between the 49 and 74 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The RAB strategy has a positive net generalized alpha for five out of the five factor models. In these cases RAB ranks between the 69 and 91 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does RAB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of RAB with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price RAB or at least to weaken the power RAB has predicting the cross-section of returns. Figure 7 plots histograms

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of RAB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RAB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RAB} RAB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RAB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on RAB. Stocks are finally grouped into five RAB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted RAB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on RAB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the RAB signal in these Fama-MacBeth regressions exceed 2.15, with the minimum t-statistic occurring when controlling for Operating leverage. Controlling for all six closely related anomalies, the t-statistic on RAB is 2.72.

Similarly, Table 5 reports results from spanning tests that regress returns to the RAB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the RAB strategy earns alphas that range from 22-30bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.15, which is achieved when controlling for Operating leverage. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the RAB trading strategy achieves an alpha of 21bps/month with a t-statistic of 3.02.

7 Does RAB add relative to the whole zoo?

Finally, we can ask how much adding RAB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the RAB signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes RAB grows to $2453.27.

8 Conclusion

This study provides compelling evidence that Residual Asset Burden (RAB) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which RAB is available.

Velikov (2023), demonstrates that RAB-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on RAB delivers an annualized Sharpe ratio of 0.39 gross (0.35 net of transaction costs), indicating strong economic significance even after accounting for implementation frictions. Most notably, the strategy generates statistically significant alphas of 25 basis points per month (t-statistic = 3.58) relative to the Fama-French five-factor model plus momentum, suggesting that RAB captures unique information about expected returns not subsumed by existing risk factors. The persistence of a 21 basis point monthly alpha (t-statistic = 3.02) after controlling for six closely related strategies—including book-to-market, leverage measures, and earnings-based ratios—further underscores RAB's incremental contribution to our understanding of the cross-section of returns.

From a practical perspective, these results suggest that RAB could serve as a valuable signal for both academic researchers and investment practitioners. The strategy's resilience to transaction costs, as evidenced by the modest difference between gross and net Sharpe ratios, indicates potential real-world applicability. Furthermore, the signal's orthogonality to existing factors suggests it could enhance portfolio construction and risk management frameworks.

However, several limitations warrant consideration. First, while our analysis follows best practices for anomaly testing, the true out-of-sample performance of RAB remains to be seen as markets evolve and adapt. Second, the economic mechanisms underlying the RAB anomaly require further theoretical and empirical investigation to understand whether it reflects systematic risk, behavioral biases, or institutional frictions. Third, our analysis focuses primarily on U.S. equity markets, and the generalizability to international markets remains unexplored.

Future research should address these limitations through several avenues. International evidence would help establish whether RAB represents a global phenomenon or is specific to U.S. market structures. Time-series analysis of the strategy's performance across different market regimes could illuminate the economic drivers of the anomaly. Additionally, examining the interaction between RAB and corporate events, such as mergers, acquisitions, or financial distress, might provide insights into the fundamental sources of the premium. Finally, investigating whether sophisticated investors trade on RAB signals through analysis of institutional holdings and trading patterns could shed light on potential limits to arbitrage that allow the anomaly to persist.

In conclusion, Residual Asset Burden emerges as a robust and economically meaningful predictor of stock returns that contributes incrementally to our understanding of asset pricing beyond existing factors. While questions remain about its theoretical underpinnings and long-term persistence, the evidence presented here suggests that RAB deserves serious consideration in both academic research and practical investment applications.

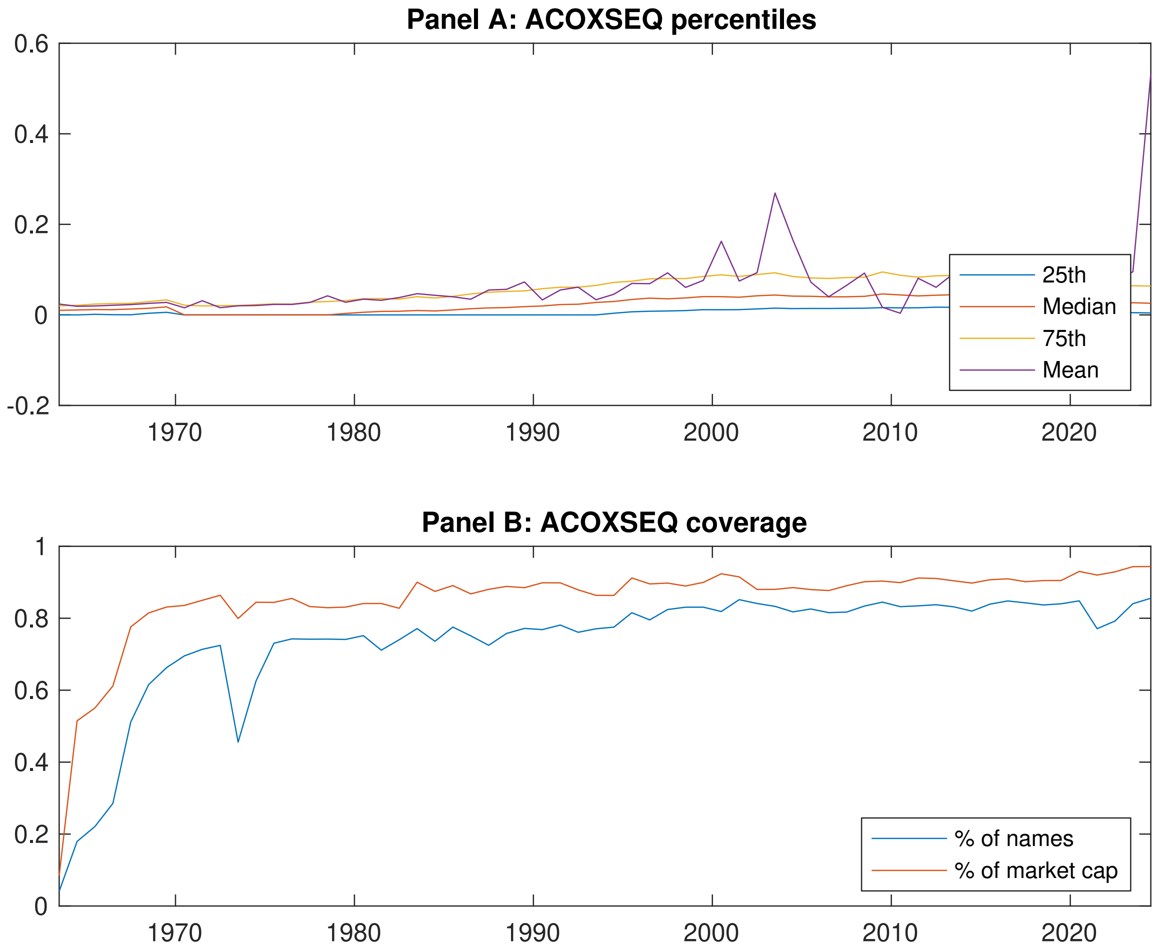


Figure 1: Times series of RAB percentiles and coverage.
This figure plots descriptive statistics for RAB. Panel A shows cross-sectional percentiles of RAB over the sample. Panel B plots the monthly coverage of RAB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on RAB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on RAB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.53	0.55	0.61	0.62	0.77	0.24
	[3.01]	[3.04]	[3.58]	[3.65]	[4.38]	[3.06]
α_{CAPM}	-0.07	-0.07	0.02	0.04	0.17	0.23
	[-1.26]	[-1.27]	[0.55]	[0.81]	[3.52]	[2.93]
α_{FF3}	-0.14	-0.05	0.05	0.09	0.21	0.35
	[-3.20]	[-0.90]	[1.09]	[2.06]	[4.46]	[5.06]
α_{FF4}	-0.12	-0.02	0.02	0.06	0.20	0.32
	[-2.71]	[-0.29]	[0.57]	[1.46]	[4.13]	[4.53]
α_{FF5}	-0.17	0.01	0.03	0.07	0.10	0.27
	[-3.62]	[0.24]	[0.59]	[1.66]	[2.40]	[3.92]
α_{FF6}	-0.15	0.04	0.01	0.05	0.10	0.25
	[-3.18]	[0.66]	[0.20]	[1.18]	[2.33]	[3.58]
Panel B: Fama and French (2018) 6-factor model loadings for RAB-sorted portfolios						
β_{MKT}	1.06	1.02	0.99	0.98	1.02	-0.04
	[93.77]	[79.40]	[92.22]	[94.45]	[96.25]	[-2.19]
β_{SMB}	-0.08	-0.01	0.05	0.00	0.08	0.17
	[-5.13]	[-0.65]	[3.51]	[0.01]	[5.49]	[6.86]
β_{HML}	0.21	-0.04	-0.08	-0.12	-0.18	-0.39
	[9.73]	[-1.67]	[-3.88]	[-5.80]	[-8.65]	[-11.91]
β_{RMW}	0.05	-0.13	0.04	0.05	0.24	0.19
	[2.10]	[-5.26]	[2.01]	[2.50]	[11.41]	[5.75]
β_{CMA}	0.05	-0.06	0.02	-0.03	0.11	0.07
	[1.49]	[-1.53]	[0.79]	[-0.99]	[3.81]	[1.39]
β_{UMD}	-0.03	-0.03	0.02	0.03	0.00	0.03
	[-2.42]	[-2.55]	[2.33]	[2.79]	[0.21]	[1.75]
Panel C: Average number of firms (n) and market capitalization (me)						
n	683	813	705	598	559	
me ($\$10^6$)	1551	1611	1899	2218	2197	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the RAB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.24	0.23	0.35	0.32	0.27	0.25
			[3.06]	[2.93]	[5.06]	[4.53]	[3.92]	[3.58]
Quintile	NYSE	EW	0.20	0.15	0.14	0.14	0.09	0.10
			[3.66]	[2.89]	[2.74]	[2.72]	[1.76]	[1.84]
Quintile	Name	VW	0.26	0.26	0.38	0.33	0.30	0.27
			[3.34]	[3.22]	[5.42]	[4.73]	[4.26]	[3.78]
Quintile	Cap	VW	0.19	0.19	0.30	0.27	0.25	0.22
			[2.56]	[2.56]	[4.62]	[4.04]	[3.68]	[3.28]
Decile	NYSE	VW	0.12	0.11	0.22	0.21	0.21	0.20
			[1.35]	[1.16]	[2.60]	[2.50]	[2.41]	[2.34]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.22	0.21	0.32	0.30	0.24	0.22
			[2.73]	[2.63]	[4.49]	[4.22]	[3.42]	[3.24]
Quintile	NYSE	EW	0.02					
			[0.35]					
Quintile	Name	VW	0.24	0.24	0.34	0.32	0.26	0.24
			[3.01]	[2.94]	[4.86]	[4.51]	[3.80]	[3.55]
Quintile	Cap	VW	0.16	0.16	0.26	0.24	0.20	0.19
			[2.21]	[2.11]	[3.91]	[3.60]	[3.01]	[2.80]
Decile	NYSE	VW	0.09	0.07	0.17	0.17	0.14	0.14
			[0.95]	[0.76]	[2.01]	[1.97]	[1.68]	[1.65]

Table 3: Conditional sort on size and RAB

This table presents results for conditional double sorts on size and RAB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on RAB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high RAB and short stocks with low RAB .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		RAB Quintiles					RAB Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.65	0.51	0.83	0.85	0.86	0.20	0.15	0.11	0.09	0.06	0.05
		[2.66]	[1.97]	[3.26]	[3.38]	[3.07]	[1.80]	[1.33]	[0.95]	[0.75]	[0.51]	[0.39]
	(2)	0.74	0.74	0.74	0.72	0.92	0.18	0.15	0.13	0.17	0.09	0.13
		[3.16]	[3.12]	[3.10]	[3.03]	[3.78]	[2.17]	[1.80]	[1.54]	[2.06]	[1.01]	[1.49]
	(3)	0.70	0.67	0.76	0.88	0.95	0.25	0.20	0.19	0.18	0.17	0.16
		[3.33]	[3.14]	[3.50]	[3.92]	[4.19]	[2.85]	[2.27]	[2.19]	[1.96]	[1.90]	[1.73]
	(4)	0.62	0.56	0.72	0.87	0.79	0.17	0.15	0.22	0.21	0.15	0.15
		[3.10]	[2.84]	[3.57]	[4.28]	[3.83]	[1.99]	[1.75]	[2.59]	[2.41]	[1.77]	[1.72]
	(5)	0.52	0.57	0.52	0.56	0.71	0.19	0.20	0.32	0.29	0.24	0.22
		[3.01]	[3.23]	[3.24]	[3.36]	[4.22]	[2.17]	[2.26]	[4.15]	[3.59]	[3.09]	[2.72]

Panel B: Portfolio average number of firms and market capitalization

		RAB Quintiles Average n					RAB Quintiles Average market capitalization (\$$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	377	385	386	386	378	29	31	33	32	29
	(2)	103	103	104	103	103	53	54	54	54	54
	(3)	72	72	72	72	72	92	90	92	92	94
	(4)	60	59	60	59	60	199	193	196	197	204
	(5)	54	54	54	54	54	1344	1401	1461	1788	1611

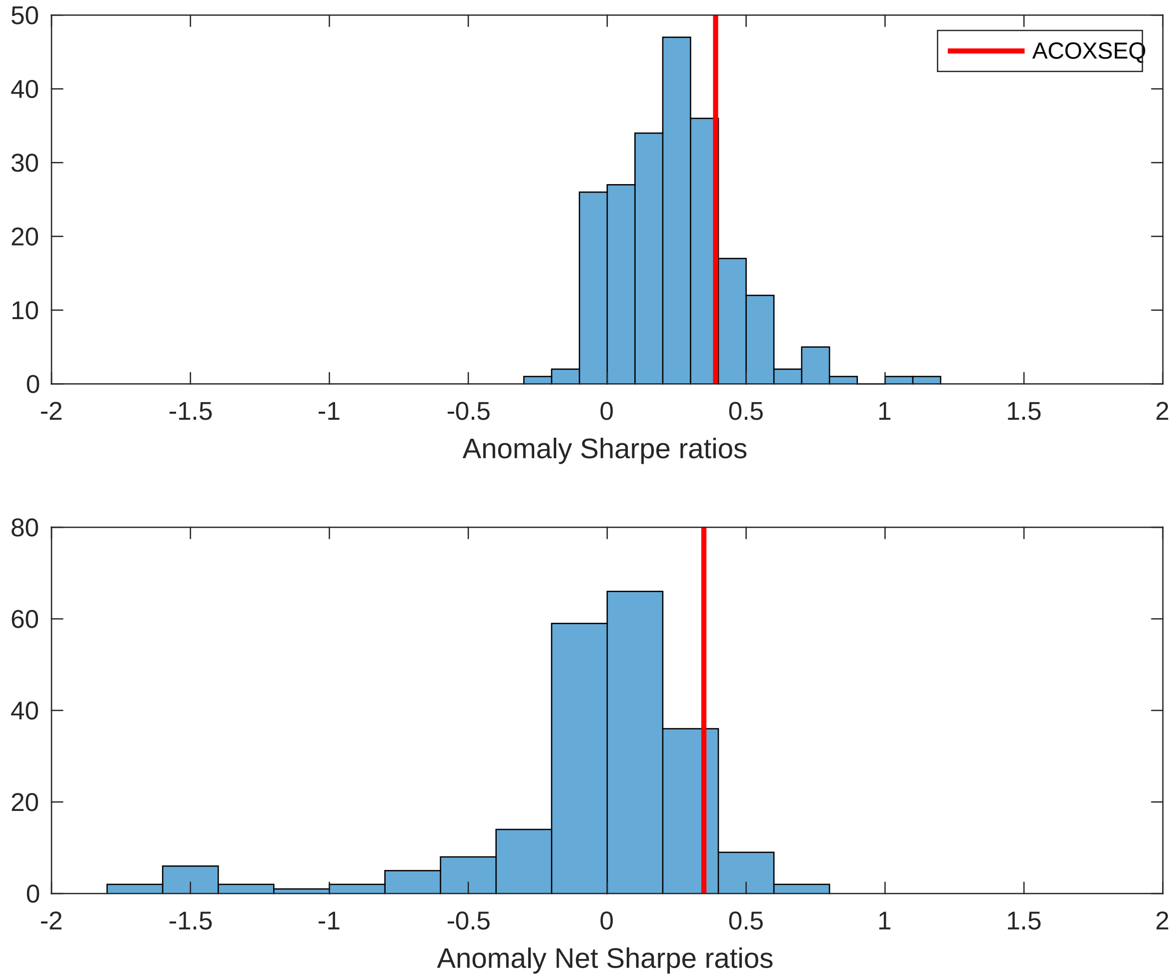


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the RAB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

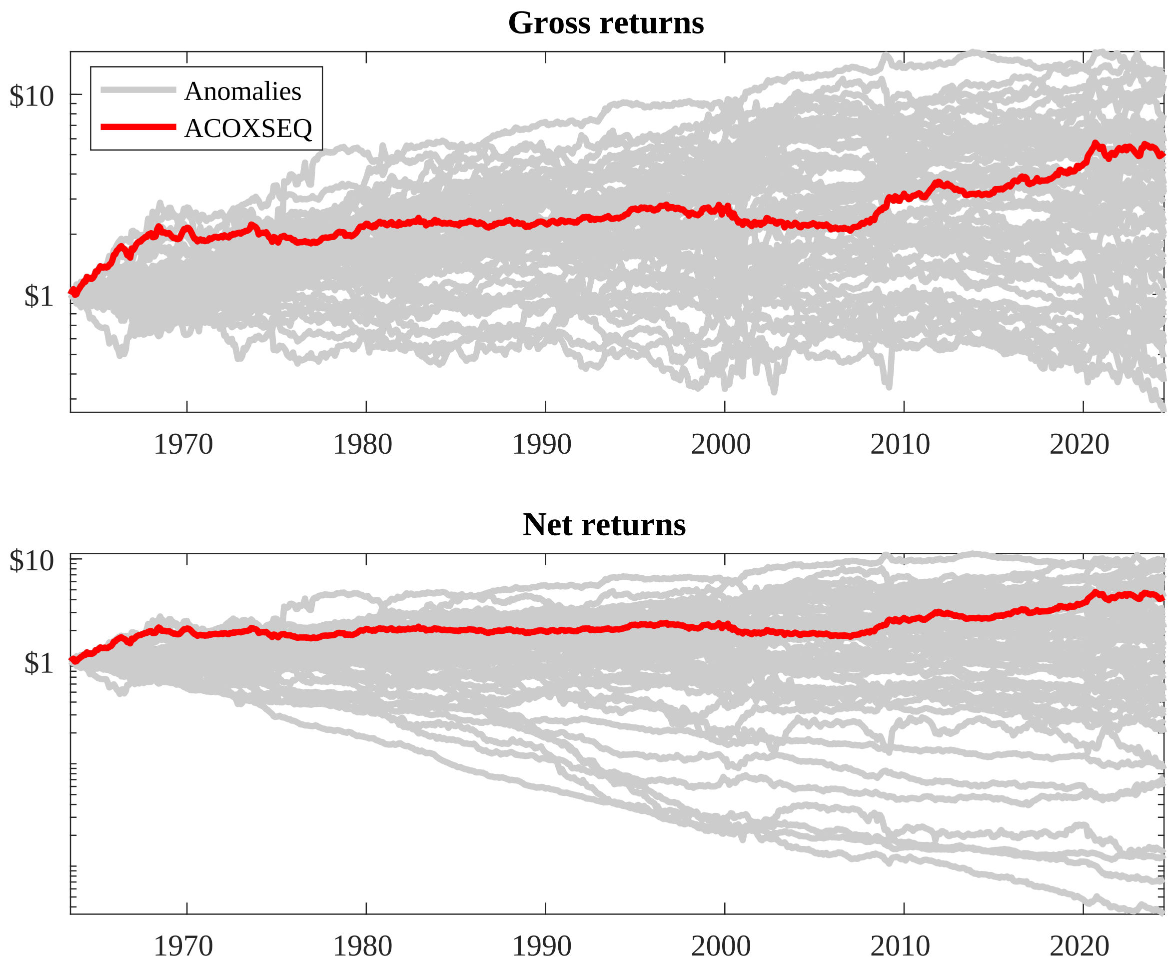
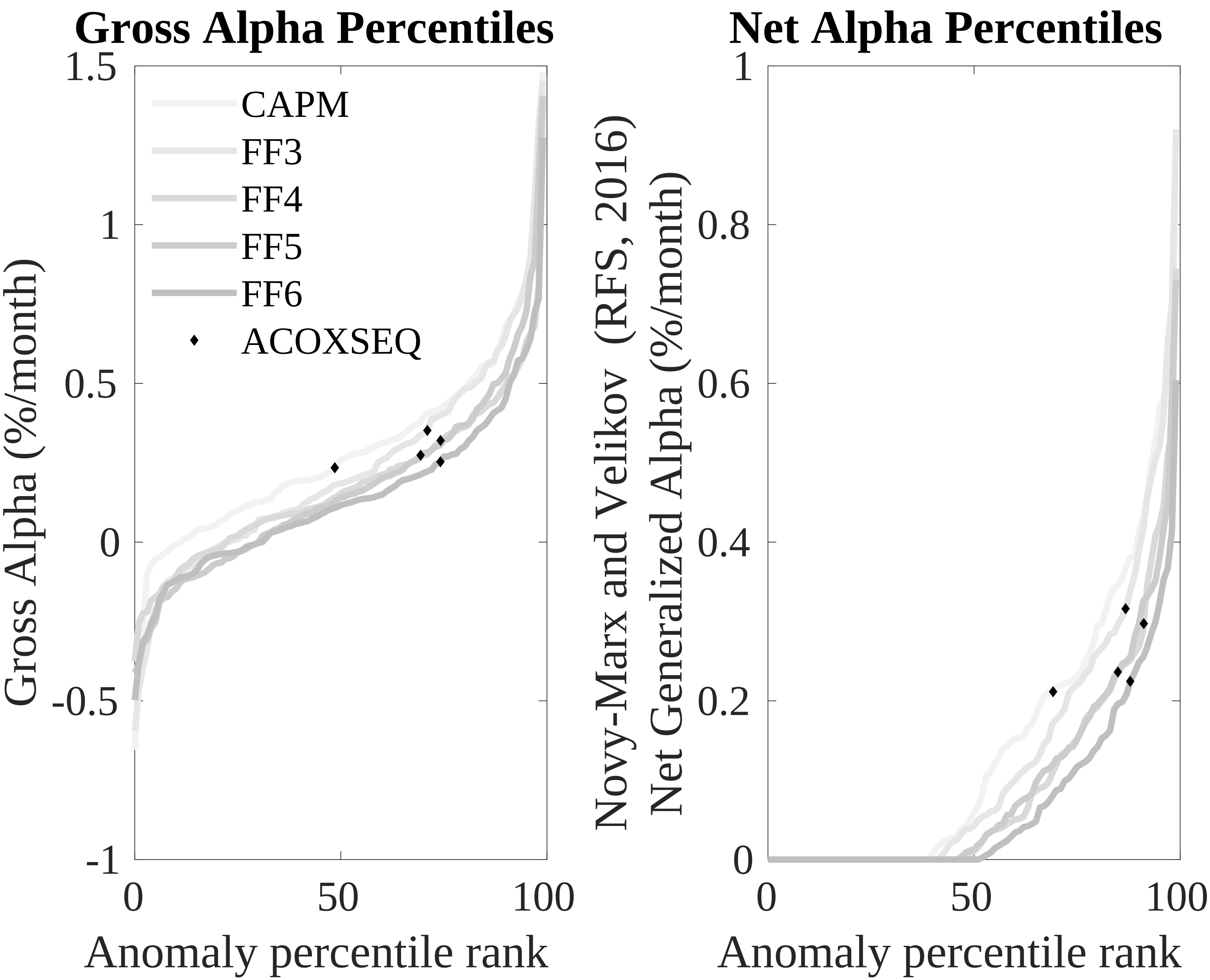
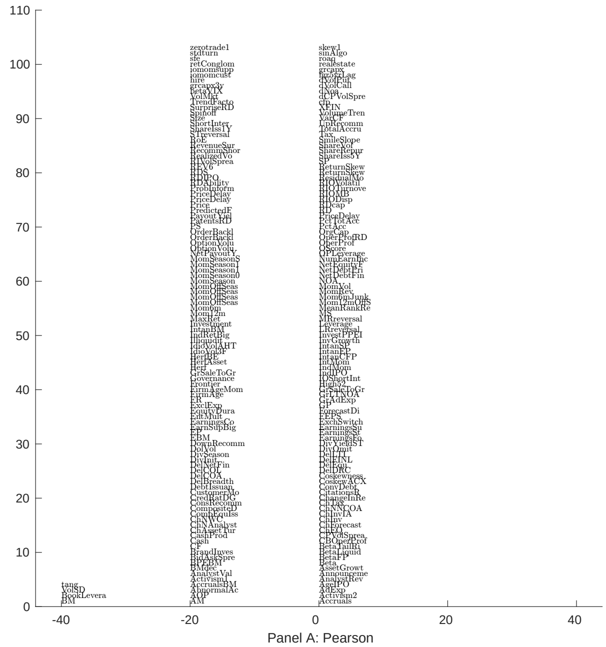


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the RAB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Net Alpha Percentiles
1.5
1
0.5
0
-0.5
-1
CAPM
FF3
FF4
FF5
FF6
ACOXSEQ
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
0.8
0.6
0.4
0.2
0
50
100
Anomaly percentile rank
Anomaly percentile rank



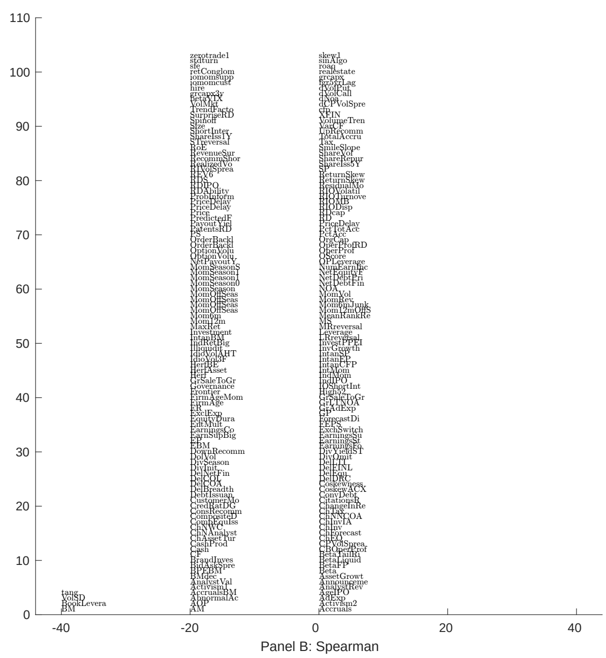


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with RAB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

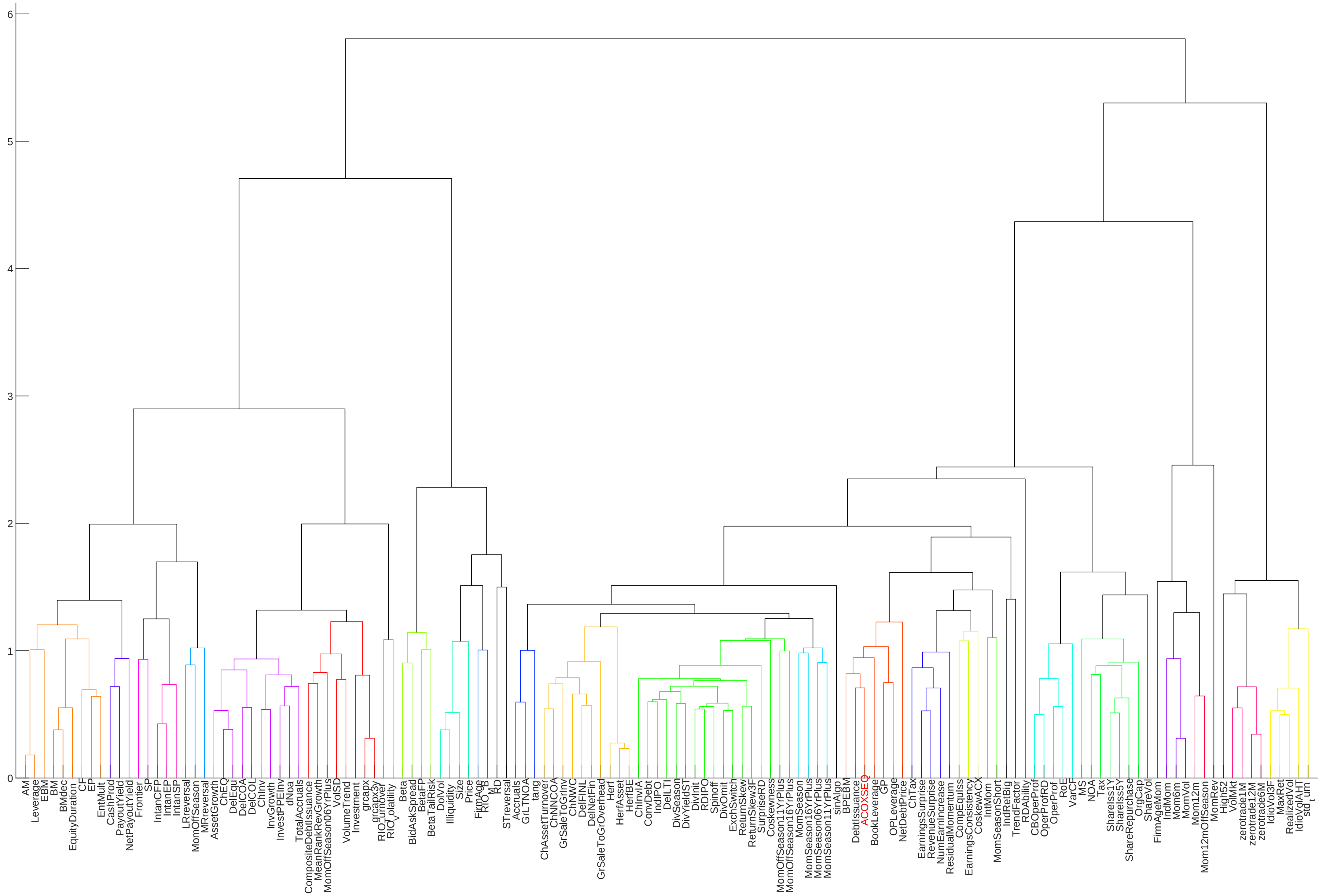


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

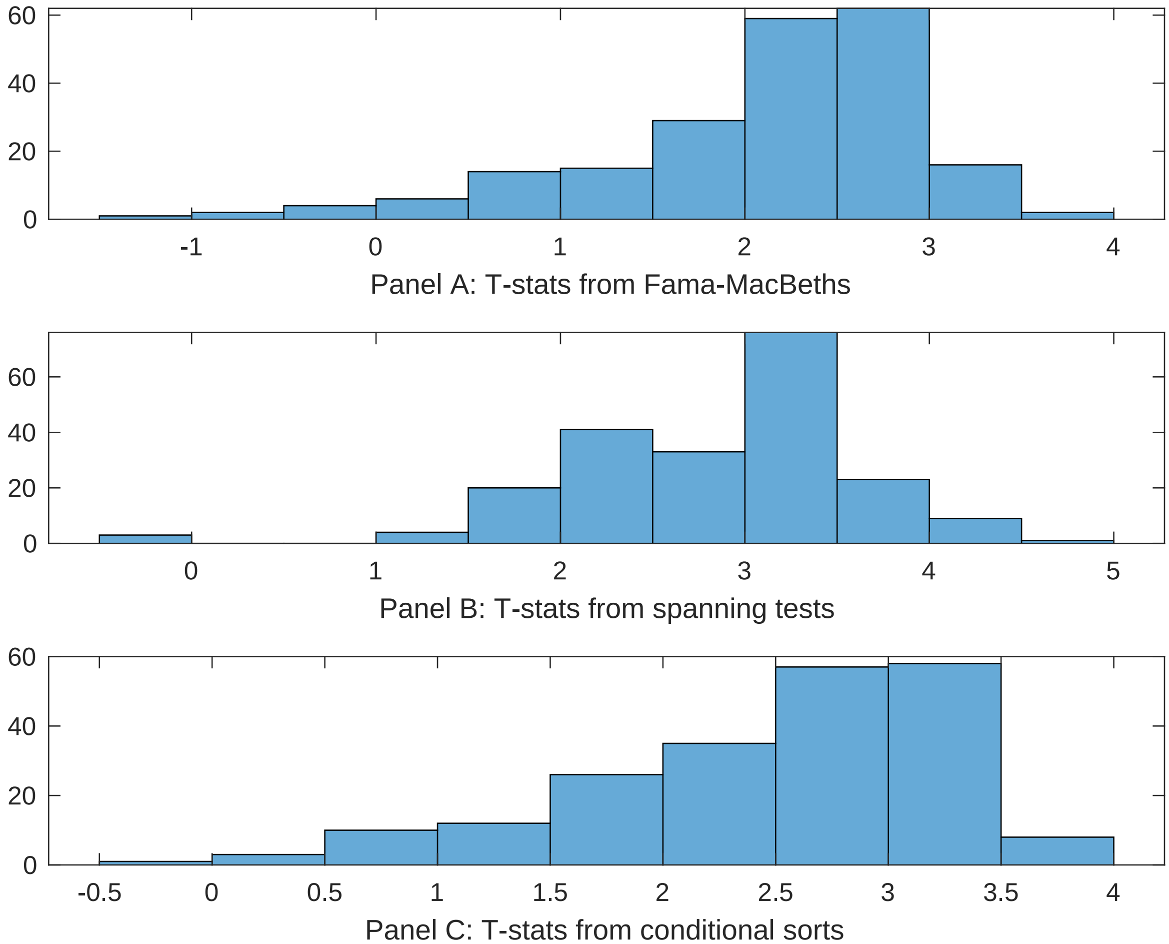


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of RAB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RAB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RAB} RAB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RAB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on RAB. Stocks are finally grouped into five RAB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted RAB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on RAB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{RAB} RAB_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Book to market, original (Stattman 1980), Book leverage (annual), Operating leverage, Equity Duration, Earnings-to-Price Ratio, Enterprise Multiple. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14	0.12	0.10	0.20	0.11	0.13	0.21
	[6.18]	[5.06]	[4.34]	[9.00]	[4.83]	[6.17]	[8.49]
RAB	0.13	0.11	0.74	0.12	0.81	0.87	0.90
	[3.43]	[3.34]	[2.15]	[3.51]	[2.46]	[2.53]	[2.72]
Anomaly 1	0.37						0.87
	[7.52]						[1.95]
Anomaly 2		0.13					-0.18
		[1.27]					[-1.23]
Anomaly 3			0.12				0.88
			[3.31]				[2.69]
Anomaly 4				0.51			0.60
				[5.30]			[5.61]
Anomaly 5					0.17		-0.59
					[2.57]		[-0.84]
Anomaly 6						0.79	0.10
						[3.41]	[3.21]
# months	715	720	720	720	715	715	715
$\bar{R}^2(\%)$	1	0	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the RAB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{RAB} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Book to market, original (Stattman 1980), Book leverage (annual), Operating leverage, Equity Duration, Earnings-to-Price Ratio, Enterprise Multiple. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.23	0.22	0.23	0.25	0.26	0.30	0.21
	[3.27]	[3.15]	[3.27]	[3.47]	[3.60]	[4.14]	[3.02]
Anomaly 1	-23.37						-19.40
	[-5.76]						[-4.19]
Anomaly 2		10.55					7.48
		[4.02]					[2.67]
Anomaly 3			19.47				16.55
			[7.03]				[5.73]
Anomaly 4				-11.77			-2.92
				[-3.23]			[-0.66]
Anomaly 5					-5.53		11.04
					[-1.76]		[2.78]
Anomaly 6						-11.31	-9.06
						[-3.91]	[-2.64]
mkt	-2.75	-2.24	-3.40	-3.45	-3.75	-4.00	-1.62
	[-1.64]	[-1.30]	[-2.06]	[-2.04]	[-2.19]	[-2.36]	[-0.97]
smb	25.75	16.46	8.35	20.48	18.23	18.98	16.52
	[9.06]	[6.75]	[3.11]	[7.70]	[7.16]	[7.62]	[5.35]
hml	-14.38	-31.20	-28.09	-25.17	-32.87	-30.24	-5.49
	[-2.75]	[-8.41]	[-8.02]	[-4.82]	[-7.40]	[-7.88]	[-0.96]
rmw	15.25	22.01	8.09	18.52	19.67	19.95	8.15
	[4.60]	[6.51]	[2.28]	[5.61]	[5.85]	[6.03]	[2.10]
cma	8.22	6.35	2.57	4.00	5.79	6.91	5.79
	[1.73]	[1.33]	[0.54]	[0.83]	[1.19]	[1.44]	[1.23]
umd	4.22	2.67	2.74	3.00	2.20	-0.86	2.07
	[2.54]	[1.59]	[1.68]	[1.79]	[1.24]	[-0.44]	[1.03]
# months	728	732	732	732	728	728	728
$\bar{R}^2$(%)	31	30	33	29	28	30	36

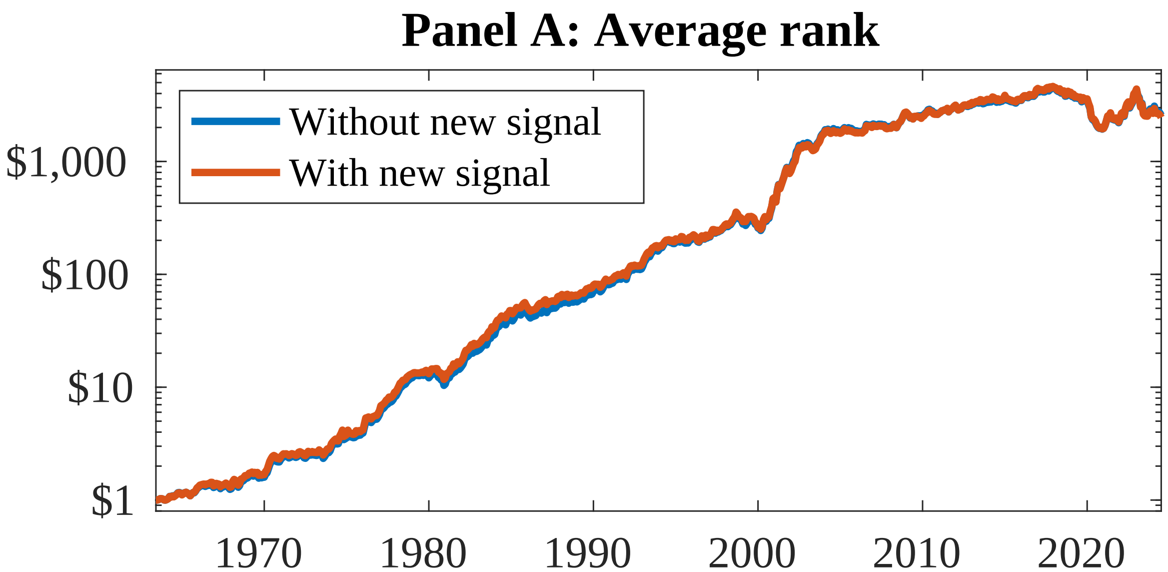


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as RAB. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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